CS 578 Statistical Machine Learning (Fall 2026) Problem Set # 5
{{Your name}} **Due: November 2nd**

Problem 1. Markov Properties.

Let $G = ([n], E)$ be a simple undirected graph, and let $X = (X_1, \ldots, X_n)$ have a joint distribution μ, not necessarily strictly positive, on finite state spaces. For $A \subseteq [n]$, write $X_A = (X_i)_{i \in A}$, and let $\partial_i = \{j \in [n] : \{i, j\} \in E\}$ denote the neighbors of vertex i. Let $U \perp V \mid W$ denote that U and V are conditionally independent given W.

Recall the following definitions from lecture:

- **Global Markov property.** For any pairwise disjoint sets $A, B, S \subseteq [n]$, if every path from a vertex in A to a vertex in B passes through S, then $X_A \perp X_B \mid X_S$.
- **Local Markov property.** For every $i \in [n]$, conditioned on its neighbors, X_i is independent of all remaining variables: $X_i \perp X_{[n] \setminus (\{i\} \cup \partial_i)} \mid X_{\partial_i}$.
- **Pairwise Markov property.** For every pair of distinct, nonadjacent vertices $i, j \in [n]$, $X_i \perp X_j \mid X_{[n] \setminus \{i,j\}}$.

1. Prove that the global Markov property implies the local Markov property.
2. Prove that the local Markov property implies the pairwise Markov property.
   Hint: you may first show that $U \perp (V, W) \mid Z \Longrightarrow U \perp V \mid (Z, W)$.

Solution.

1. Fix a vertex i, and take

$$A = \{i\}, \qquad S = \partial_i, \qquad B = [n] \setminus (\{i\} \cup \partial_i).$$

   These sets are pairwise disjoint. Every path from i to a vertex of B must first visit a neighbor of i, hence must pass through S. The global Markov property therefore gives $X_i \perp X_B \mid X_{\partial_i}$, which is precisely the local Markov property.

2. First prove the weak union implication in the hint. Suppose $U \perp (V, W) \mid Z$, and fix (z, w) with $\Pr(Z = z, W = w) > 0$. Conditional independence gives

$$\begin{aligned} \Pr(U = u, V = v \mid Z = z, W = w) &= \frac{\Pr(U = u, V = v, W = w \mid Z = z)}{\Pr(W = w \mid Z = z)} \\ &= \Pr(U = u \mid Z = z) \Pr(V = v \mid Z = z, W = w). \end{aligned}$$

   Summing over v shows that $\Pr(U = u \mid Z = z, W = w) = \Pr(U = u \mid Z = z)$. Substituting this into the factorization above proves $U \perp V \mid (Z, W)$.

Now let i, j be distinct nonadjacent vertices and set $R = [n] \setminus (\{i, j\} \cup \partial_i)$. Since $j \notin \partial_i$, the local Markov property at i is

$$X_i \perp (X_j, X_R) \mid X_{\partial_i}.$$

Since $\partial_i \cup R = [n] \setminus \{i, j\}$, weak union yields

$$X_i \perp X_j \mid X_{[n]\setminus\{i,j\}}.$$

This is the pairwise Markov property. All conditional probabilities used above condition on positive-probability events, so strict positivity of μ is unnecessary.

Problem 2. Higher-order Tensor Decomposition.

Let $u \odot v$ denote the Khatri–Rao product of two vectors: if $u \in \mathbb{R}^m$ and $v \in \mathbb{R}^n$, then

$$u \odot v = \mathrm{vec}(uv^\top) \in \mathbb{R}^{mn},$$

where vec flattens a matrix column by column.

The *Kruskal rank* (or k-rank) of a collection of vectors $u_1, u_2, \ldots, u_m \in \mathbb{R}^n$ is the largest $k \in \{0, \ldots, m\}$ such that every subset of k vectors is linearly independent.

In this problem, we explore properties of the Khatri–Rao product and use them to design algorithms for decomposing higher-order tensors.

(a) Let k_u and k_v be the k-ranks of $u_1, u_2, \ldots, u_m$ and $v_1, v_2, \ldots, v_m$, respectively. Assume all u_i and v_i are nonzero. Prove that the k-rank of

$$u_1 \odot v_1,\ u_2 \odot v_2,\ \ldots,\ u_m \odot v_m$$

is at least $\min(k_u + k_v - 1, m)$.

(b) Construct a family of examples where the k-rank of

$$u_1 \odot u_1,\ u_2 \odot u_2,\ \ldots,\ u_m \odot u_m$$

is exactly $2k_u - 1$, and not any larger. To make this nontrivial, your examples must satisfy $m > 2k_u - 1$.

(c) Given an $n \times n \times n \times n \times n$ fifth-order tensor

$$T = \sum_{i=1}^{r} a_i^{\otimes 5},$$

give an algorithm for finding its factors that works for $r = 2n - 1$, under appropriate conditions on the factors $a_1, a_2, \ldots, a_r \in \mathbb{R}^n$.

Hint: Reduce to the third-order case.

Solution.

(a) Set $K = \min(k_u + k_v - 1, m)$. Since all factors are nonzero, $k_u, k_v \geq 1$. Fix any $S \subseteq [m]$ with $|S| = s \leq K$, and suppose

$$\sum_{j \in S} c_j (u_j \odot v_j) = 0, \qquad \text{equivalently} \qquad \sum_{j \in S} c_j u_j v_j^\top = 0.$$

Fix $i \in S$. Choose $J \subseteq S \setminus \{i\}$ with $|J| = \min(k_u - 1, s - 1)$. The vectors indexed by $J \cup \{i\}$ are linearly independent, so there is a linear functional ℓ such that $\ell(u_i) = 1$ and $\ell(u_j) = 0$ for $j \in J$. Applying ℓ to the first factor of the matrix identity gives

$$\sum_{j \in S \setminus J} c_j \ell(u_j) v_j = 0.$$

There are

$$|S \setminus J| = \max(1, s - k_u + 1) \le k_v$$

remaining vectors, so they are linearly independent. In particular, $c_i \ell(u_i) = c_i = 0$. Since i was arbitrary, all coefficients vanish. Thus every subset of at most K product vectors is independent, proving the claimed lower bound.

(b) For any integer $k \ge 1$ and any $m \ge 2k$, choose distinct real numbers $t_1, \dots, t_m$, and define

$$u_i = (1, t_i, t_i^2, \dots, t_i^{k-1})^\top \in \mathbb{R}^k.$$

Any k of these vectors form an invertible Vandermonde matrix, so $k_u = k$. For $0 \le d \le 2k-2$, let H_d be the $k \times k$ matrix whose (p,q)-entry is 1 when $p+q-2 = d$, and 0 otherwise. Then

$$u_i \odot u_i = \sum_{d=0}^{2k-2} t_i^d \operatorname{vec}(H_d).$$

Hence all product vectors lie in a subspace of dimension at most $2k-1$. Because $m \ge 2k$, every set of $2k$ of them is dependent, giving k-rank at most $2k-1$. Part (a) gives the matching lower bound $\min(2k-1, m) = 2k-1$. These examples therefore have exactly the required k-rank, with $m > 2k_u - 1$.

(c) Assume first that $n \ge 2$ and that the factors have k-rank n; that is, every n of the a_i are linearly independent. This condition holds for generic factors. Write

$$r = 2n-1, \qquad b_i = a_i \odot a_i, \qquad B = [b_1 \ \cdots \ b_r] \in \mathbb{R}^{n^2 \times r}.$$

By part (a), the k-rank of B is at least $\min(2n-1, r) = r$. Thus B has full column rank.

Step 1: Group the modes. Group modes $(1,2)$, $(3,4)$, and 5, using the column-by-column ordering of vec. This reshapes T into the third-order tensor

$$G = \sum_{i=1}^{r} b_i \otimes b_i \otimes a_i \in \mathbb{R}^{n^2 \times n^2 \times n}.$$

This is a third-order decomposition with two full-column-rank factor matrices, so it can be recovered by simultaneous diagonalization as follows.

Step 2: Recover the directions of the b_i. Choose independent standard Gaussian vectors $x, y \in \mathbb{R}^n$. Contract the last mode to form

$$M_z = G(\cdot, \cdot, z) = B D_z B^\top, \qquad D_z = \operatorname{diag}(a_1^\top z, \dots, a_r^\top z), \quad z \in \{x, y\}.$$

With probability one, all $a_i^\top y$ are nonzero and the ratios

$$\rho_i = \frac{a_i^\top x}{a_i^\top y}, \qquad 1 \le i \le r,$$

are pairwise distinct. Indeed, the k-rank assumption implies that every a_i is nonzero and no two are proportional. Each forbidden equality is therefore the zero set of a nonzero polynomial in (x, y), and there are only finitely many such equalities.

Compute an orthonormal basis $Q \in \mathbb{R}^{n^2 \times r}$ for $\mathrm{range}(M_y) = \mathrm{range}(B)$. Let $C = Q^\top B$, which is invertible. The computable $r \times r$ matrix

$$\begin{aligned} K &= (Q^\top M_x Q)(Q^\top M_y Q)^{-1} \\ &= (CD_xC^\top)(CD_yC^\top)^{-1} = C\,\mathrm{diag}(\rho_1, \ldots, \rho_r)C^{-1} \end{aligned}$$

has distinct eigenvalues. Its eigenvectors are the columns of C, up to nonzero scaling and permutation. Multiplying these eigenvectors by Q therefore recovers vectors $\widehat{b}_i = s_i b_i$, with $s_i \neq 0$, in some order. If M_y has rank less than r or K has repeated eigenvalues, resample x, y.

Step 3: Recover the factors, including their scales. Reshape each $\widehat{b}_i$ into an $n \times n$ matrix:

$$\mathrm{unvec}(\widehat{b}_i) = s_i a_i a_i^\top.$$

Any nonzero column is a vector $d_i = t_i a_i$ for some nonzero real t_i. Solve the linear system in the r scalar unknowns λ_i:

$$T = \sum_{i=1}^{r} \lambda_i d_i^{\otimes 5}.$$

A solution exists, namely $\lambda_i = t_i^{-5}$. It is unique: the vectors $d_i^{\otimes 2} = t_i^2 b_i$ are independent, so there are linear functionals L_j with $L_j(d_i^{\otimes 2}) = \delta_{ji}$. Applying L_j to the first two modes of a relation $\sum_i \gamma_i d_i^{\otimes 5} = 0$ gives $\gamma_j d_j^{\otimes 3} = 0$, forcing $\gamma_j = 0$. Finally output

$$\widehat{a}_i = \sqrt[5]{\lambda_i}\, d_i,$$

taking the unique real fifth root. Since $\sqrt[5]{t_i^{-5}} = t_i^{-1}$, this recovers every original a_i exactly, up to permutation. Thus the algorithm succeeds with probability one in exact arithmetic under the stated k-rank condition, even though $r = 2n - 1 > n$.

If $n = 1$, then $r = 1$ and the tensor has a single entry; its real fifth root is the required scalar factor.